CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Purple Company E-Commerce Database System is designed to efficiently manage customers, product categories, products, orders, order items, payments, deliveries, suppliers, and administrators. Functional dependencies identify the relationship between attributes in each table, while redundancy analysis and data anomaly detection ensure data consistency and minimize duplication. A properly normalized database improves performance, maintains data integrity, and supports efficient business operations.

1. CUSTOMER TABLE

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_Number, Address

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders reference customers using Customer_ID, avoiding duplicate customer records.

Data Anomalies

Insert Anomaly: A new customer can be added without placing an order.

Update Anomaly: Updating customer details requires changes only in the Customer table.

Delete Anomaly: Deleting an order does not remove customer information.

2. CATEGORY TABLE

Functional Dependency

Category_ID → Category_Name

Redundancy Analysis

Each category is stored once. Products reference the category using Category_ID instead of repeating the category name.

Data Anomalies

Insert Anomaly: New categories can be added without products.

Update Anomaly: Category name is updated only once.

Delete Anomaly: Deleting a product does not delete the category.

3. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Brand, Price, Stock, Category_ID

Redundancy Analysis

Product information is stored only in the Product table. Order items reference products using Product_ID, avoiding repeated product details.

Data Anomalies

Insert Anomaly: Products can be added before any orders are placed.

Update Anomaly: Product price or stock is updated only once.

Delete Anomaly: Deleting an order does not remove product information.

4. ORDERS TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

The Orders table stores only order-related information. Customer details are referenced using Customer_ID.

Data Anomalies

Insert Anomaly: Orders can be created after selecting an existing customer.

Update Anomaly: Order status is updated without affecting other tables.

Delete Anomaly: Deleting an order does not remove customer information.

5. ORDER_ITEM TABLE

Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

This table links orders and products, eliminating repeated product information in the Orders table.

Data Anomalies

Insert Anomaly: Products can be added to an existing order independently.

Update Anomaly: Updating quantity affects only the selected order item.

Delete Anomaly: Removing an order item does not delete the product.

6. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status

Redundancy Analysis

Payment information is maintained separately from Orders, preventing duplicate payment records.

Data Anomalies

Insert Anomaly: Payment can be recorded after an order is created.

Update Anomaly: Payment status is updated only once.

Delete Anomaly: Deleting payment information does not remove the related order.

7. DELIVERY TABLE

Functional Dependency

Delivery_ID → Order_ID, Tracking_Number, Delivery_Status

Redundancy Analysis

Delivery information is stored separately, preventing repeated shipment details.

Data Anomalies

Insert Anomaly: Delivery records can be created after order confirmation.

Update Anomaly: Delivery status is updated independently.

Delete Anomaly: Deleting delivery information does not affect order records.

8. SUPPLIER TABLE

Functional Dependency

Supplier_ID → Supplier_Name, Contact_Number, Email

Redundancy Analysis

Supplier information is stored only once and can be referenced whenever required, avoiding duplicate supplier records.

Data Anomalies

Insert Anomaly: New suppliers can be added without products.

Update Anomaly: Supplier details are updated only once.

Delete Anomaly: Deleting a supplier does not affect existing order records.

9. ADMIN TABLE

Functional Dependency

Admin_ID → Admin_Name, Email, Password

Redundancy Analysis

Administrator information is maintained separately to avoid duplication of login details.

Data Anomalies

Insert Anomaly: New administrators can be added independently.

Update Anomaly: Admin details are updated only once.

Delete Anomaly: Removing an administrator does not affect customer, product, or order information.

CONCLUSION

The Purple Company E-Commerce Database System is designed using proper functional dependencies and normalized tables. Each entity stores only relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design ensures data consistency, maintains integrity, and supports efficient database management for the Purple Company.